

What Is Learned in Machine Learning?

Authors: Yousuf Mazahim and John-Paul Greenman

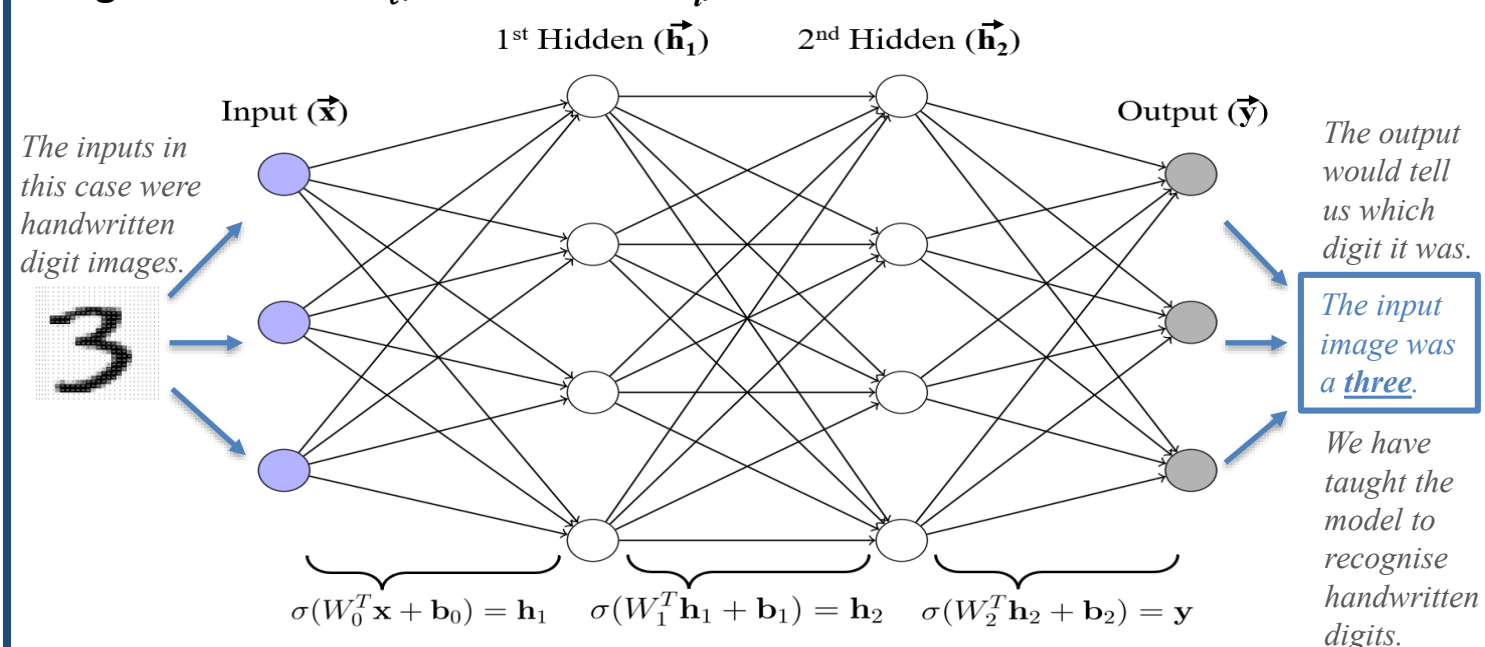
Supervisor: Dr Igor Smolyarenko

Motivation

Models help us explain the world around us and make predictions. For example, **Newton's laws of motion** form a model of how objects move: $F = ma$ (force equals mass times acceleration).

Like Newton's equations, **machine learning models** are also mathematical systems. However, instead of being engineered by scientists, machine learning models **learn patterns from data**.

A powerful machine learning model is the **deep neural network (DNN)**. Neural networks can learn complex nonlinear patterns from data and are used in technologies like **ChatGPT**. The learned information is stored matrices and vectors called **weights** and **biases**. Below, we have a four-layer DNN showing the transformations from one layer to the next using weight matrices W_i , bias vectors b_i , and activation functions σ .



This project studies the **structure of the weight matrices** between hidden layers (like W_1) in neural networks after being trained on data. We test if **learned information** can be **isolated** i.e. $W_1 = W_{\text{info}} + W_{\text{noise}}$. This has great implications on **efficiency** and **energy** as less information needs to be stored.

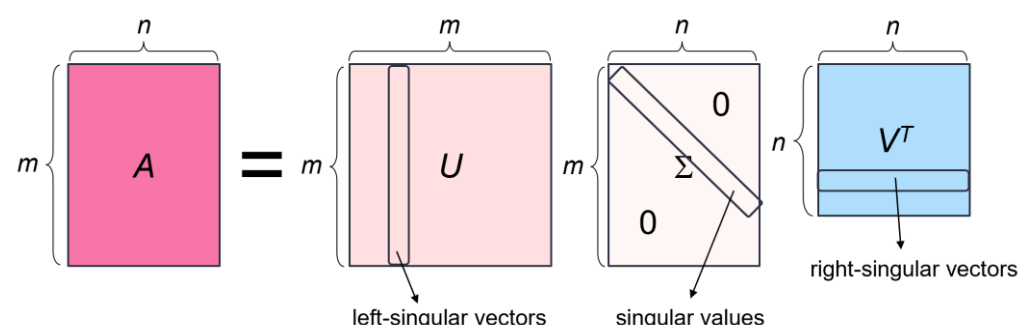
Methodology

To understand the structure of these hidden weight matrices, we used tools from **linear algebra**, **statistics** and **random matrix theory**. These are:

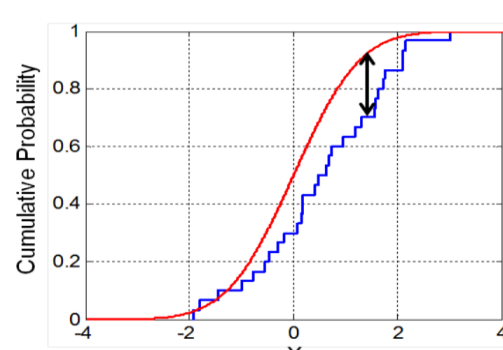
- **Singular Value Decomposition (SVD)**: to break down weight matrices into learned information and whatever remains (we show that these are like matrices with randomly generated entries i.e. random noise).
- **Kolmogorov-Smirnov testing**: to compare distributions from our weight matrices against those from random noise matrices.
- **Random Matrix Theory (RMT)**: to define what "random noise" looks like using known limiting distributions.

Related to SVD, the **rank** of a matrix tells us how much **useful** information is stored. A low-rank matrix would have more repeated patterns or **correlated** rows/columns. A full-rank matrix would have little redundancy in it i.e. you **cannot compress** it.

When you apply **SVD** on any matrix, you obtain **its left and right singular vectors** and their corresponding **singular values**. The number of non-zero singular values is equal to the rank. Each singular value shows how much of the original data can be explained by one simplified layer of structure i.e. bigger values mean **clearer patterns**, smaller ones are like **background noise**.

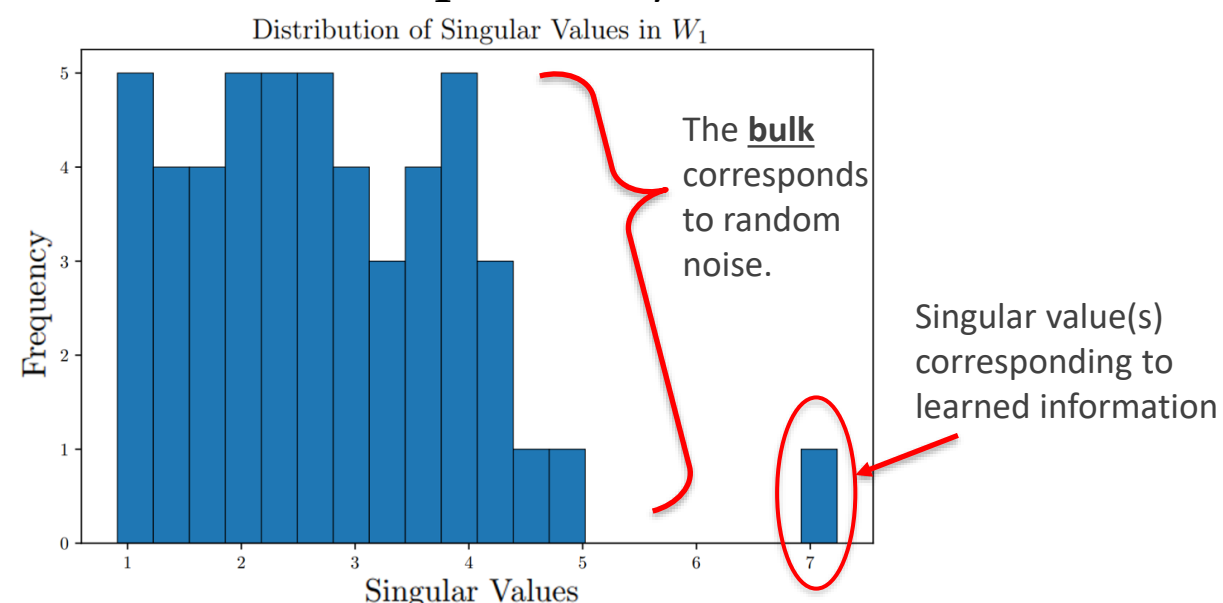


The **Kolmogorov-Smirnov test** checks whether two distributions are **similar** by comparing their cumulative curves. In the plot, the red curve is what we **expect** (e.g. a theoretical model), and the blue steps show what we **observed** in our data. The KS test measures the **biggest gap** between these two curves (shown by the black arrow). If the gap is small, the data likely came from the model. If the gap is large, it probably didn't.



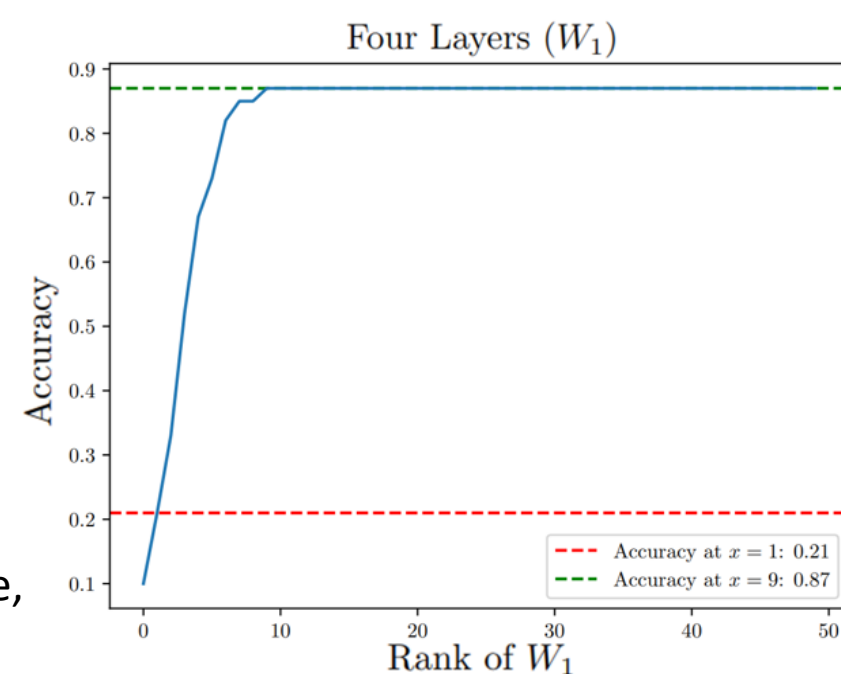
Key Findings

We used Singular Value Decomposition to study how trained neural networks store information in their weight matrices. In **classification networks** (their output is one from a finite number of **classes**), we found that only a **small number** of large singular values carry learned structure, while the rest resemble random noise. We then tested how consistent this pattern is across networks of different sizes and purposes. Notice the SVD of W_1 in a binary classification model.

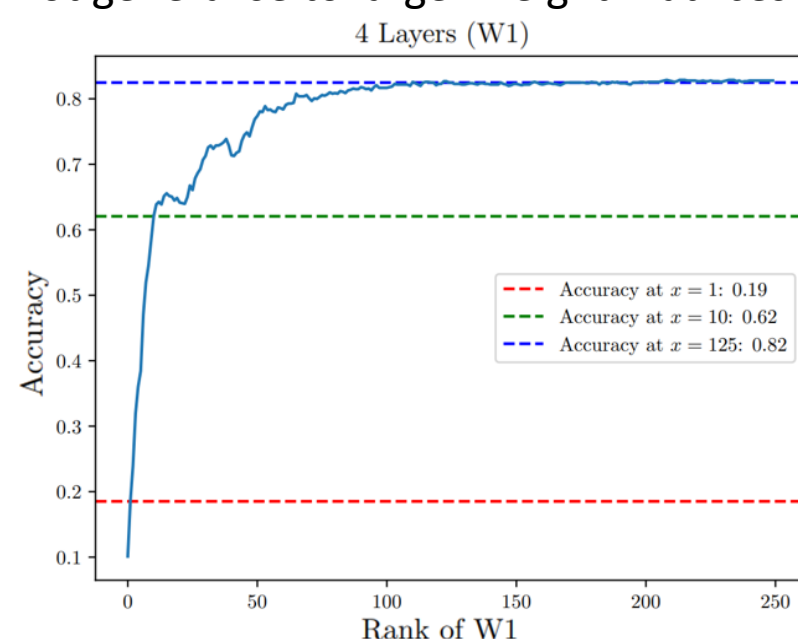


We verified this by setting the **bulk** of singular values we believed represented random noise to **zero** and reconstructing the weight matrix. The rank of W_1 is defined as the number of its non-zero singular values. We see below in a different ten-class model that we only need **nine** out of the available fifty ranks of W_1 to achieve the best accuracy where accuracy is the percentage of correctly predicted digits.

The logical next step is to verify what we hypothesised to be random noise. After using the **KS-test** to compare the matrix made from the bulk singular values to the **Porter-Thomas** distribution which models random noise, we confirm the hypothesis, $W_1 = W_{\text{info}} + W_{\text{noise}}$ for the ten-class and binary classification networks.



This number corresponds to the size of our output (the number of classes). However, this does not generalise to larger weight matrices. We believe this is because larger weight matrices have a **larger bulk** of singular values. We say the larger bulk of singular values **masks** the outlier corresponding to learned information, meaning we are unable to isolate it.



In collaboration with **John-Paul Greenman**, we extended our analysis to a neural network trained to solve **ordinary differential equations (ODEs)**. This means the output of the network is now continuous. As a result, **almost full rank** is required for minimal error. We believe it may be because of a **sparse spread** of the singular values corresponding to learned information or the **masking** in larger matrices. This is **interesting** as random noise structure predicts well so it requires further research.

